Architecture-Dependent Causal Transfer of Activation States Across Large Language Models

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Abstract

Direct communication between artificial intelligence systems currently relies on natural language as an intermediate layer, incurring encoding/decoding overhead, token cost, and latency. We ask whether internal activation states can instead be transferred causally between different large language model (LLM) architectures via a learned projection, and we evaluate this question at three levels: representational similarity, cross-model retrieval from projected states, and end-to-end causal transfer via activation injection during generation. Using four architecturally diverse open-weight models (Qwen2-0.5B, Phi-3-mini, Mistral-7B, FLAN-T5-base), we find that representational alignment in trained models exceeds a random-initialization null baseline and is best captured by a rank-based metric (mutual k-nearest-neighbour alignment), which is more robust to activation-magnitude outliers than centered kernel alignment (CKA) or Procrustes analysis. A learned projection network retrieves the correct target-model representation from a held-out set well above chance for the three causal decoder-only model pairs (45–50% top-1 accuracy vs. 5% chance) but at chance level for the encoder-based FLAN-T5. Injecting projected activations into a target model during generation, however, produces a statistically significant, pre-registered causal effect on retrieval-based output similarity for only one of the three decoder-only model pairs (Qwen2-0.5B→Phi-3-mini: 23.3% vs. 0.0% under negative control, $p = 0.047$, false-discovery-rate corrected); the two model pairs targeting Mistral-7B show no such effect despite comparable representational alignment at the hidden-state level. We interpret these results as evidence for causal transfer of the representational *vehicle*, not of meaning, and conclude that end-to-end activation-state transfer between LLMs, as currently implemented, is architecture-dependent rather than universal.

1 Introduction

Communication between artificial intelligence systems today runs almost exclusively through natural language: one model's output is decoded into text, transmitted, and re-encoded by another model. This intermediate layer is a plausible source of overhead — encoding and decoding cost, token consumption, latency, and potential information loss in translation — that would not be incurred if models could instead exchange internal representations directly. Whether such direct exchange is feasible depends on a prior, largely open empirical question: do different LLM architectures, trained independently on different data by different organisations, encode information in representational spaces that are similar enough to support a learned mapping between them?

A growing body of work suggests that internal representations of neural networks trained on related tasks or data are more similar than one might expect a priori, and a range of alignment metrics — centered kernel alignment [Kornblith et al., 2019], neuron-level correlation [Ehsani Oskouie et al., 2024], and structural compatibility frameworks for representations shaped by different architectural constraints [Nikooroo and Engel, 2025] — has been developed to quantify this similarity. At the strongest end of this line of work, the Platonic Representation Hypothesis proposes that representations of sufficiently large models trained on sufficiently broad data converge toward a shared statistical model of the world, largely independent of architecture or modality [Huh et al., 2024]. These findings establish that representations can be *similar*; they do not by themselves establish that a similarity signal can be *exploited* — projected onto a target model and shown to causally change its behaviour in a controlled, verifiable way — nor do they license any claim about what, if anything, such a transfer would mean for the target model.

This paper reports three linked experiments addressing that gap, run under pre-registered success criteria fixed before each test was executed: (1) an extended representational-similarity analysis across four architecturally diverse open-weight LLMs, using both raw and rank-based alignment metrics together with random-initialization null baselines and a lexical-triviality control; (2) a cross-model projection network evaluated by a pre-registered top-1 retrieval-accuracy metric on data disjoint from the similarity analysis; and (3) an end-to-end causal state-transfer test that injects a projected activation vector into a target model during generation and evaluates the effect against a negative-control injection. We report all three sets of results as obtained, including model pairs for which the pre-registered success criteria were not met. We further scope our central causal claim explicitly to the transfer of a representational *vehicle* — an activation pattern with a demonstrable causal effect on downstream output — rather than to the transfer of meaning or shared understanding (Section 5.1), independent of which model pairs the end-to-end test succeeded for.

2 Related Work

2.1 Cross-model representational alignment

Centered kernel alignment (CKA) was introduced as a similarity index for comparing neural network representations that, unlike earlier canonical-correlation-based measures, remains reliable in high-dimensional, low-sample regimes and can recover correspondences between representations from different random initialisations [Kornblith et al., 2019]. Subsequent work has extended cross-model comparison along two directions relevant here: neuron-level correlation methods that relate a new model's internal structure to a reference model to predict performance and generalisation [Ehsani Oskouie et al., 2024], and structural frameworks that explicitly model how architectural constraints ("shaping operators") affect the compatibility of representations learned under different inductive biases [Nikooroo and Engel, 2025]. A separate methodological line shows that superposition — multiple features encoded in overlapping neuron directions — can distort standard alignment metrics and must be disentangled before those metrics can be interpreted as measuring "true" representational alignment [Longon et al., 2025]; we return to this point in Section 4, where it directly affects our choice of primary metric.

2.2 The Platonic Representation Hypothesis as a competing explanation

The Platonic Representation Hypothesis holds that representations across models, architectures, and even modalities converge toward a shared statistical model of the world as scale and data diversity increase [Huh et al., 2024]. This hypothesis is a natural competing explanation for any representational similarity we observe: if alignment between models reflects generic, architecture-independent convergence driven by shared task structure and training-data statistics, it would not indicate a model-specific, exploitable channel so much as a general property of sufficiently trained networks. We treat this as a hypothesis to be weighed against our data rather than as a citation to be discharged in passing, and return to it in Section 5 in light of two of our own findings: that architectural family (causal decoder-only vs. bidirectional encoder attention) predicts cross-model alignment more strongly than company or training-data origin does, and that a lexically disguised synonymy signal — a harder test of shared meaning than mere topical association — is robust in only one of the four models tested.

2.3 Vehicle, content, and symbol grounding

A separate literature in philosophy of mind distinguishes the *vehicle* of a representational state — the physical or computational carrier that realises it — from its *content* — what it represents [Hurley, 1998]. This distinction is closely related to the symbol grounding problem: a formal symbol system's internal consistency does not by itself establish that its symbols are grounded in, i.e. connected to, anything external to the system [Harnad, 1990]. Both ideas are directly relevant to interpreting any successful cross-model state transfer, and we adopt them as the conceptual basis for the scope of our own causal claim in Section 5.1.

3 Methods

3.1 Models

We used four open-weight LLMs spanning three independent developers and two architectural families: Qwen2-0.5B [Yang et al., 2024], Phi-3-mini-128k- instruct in 4-bit quantisation [Abdin et al., 2024], Mistral-7B-v0.1 in 4-bit quantisation [Jiang et al., 2023], and FLAN-T5-base, an encoder-decoder model combining the T5 architecture and pre-training objective [Raffel et al., 2020] with the Flan instruction-tuning procedure [Chung et al., 2022]. Qwen2, Phi-3, and Mistral are causal decoder-only transformers; FLAN-T5's encoder, from which we extract representations, uses bidirectional attention. This selection was chosen to vary both training organisation (Alibaba, Microsoft, Mistral AI, Google) and attention architecture (causal vs. bidirectional), so that the two possible sources of representational similarity — shared architecture and shared/overlapping training data or objectives — could, to a first approximation, be told apart.

3.2 Representational similarity and alignment metrics

We compared hidden-state representations for 12 RELATED concept pairs (topically/associatively linked, lexically dissimilar) and 12 SYNONYM concept pairs (semantically near-identical, lexically dissimilar) across models and layers. The SYNONYM set is the harder triviality control: because its members are lexically dissimilar, any alignment signal it produces cannot be explained by shared surface tokens, unlike a naive Layer-0/1 lexical-overlap artifact. Representations were pooled by mean-pooling token embeddings while excluding special tokens; for causal models we additionally excluded token position 0. This exclusion was necessary because an early debugging phase found that, for causal models, naive mean-pooling over all tokens produced near-ceiling cosine similarity (≈ 1.000) for essentially any word pair, including random pairs, driven by a single early-position token whose activation dominates the pooled vector regardless of content. This is consistent with two independently reported phenomena in causal decoder-only transformers: a small number of disproportionately large-magnitude activations concentrated at fixed positions, including the first token [Sun et al., 2024], and anomalously high, largely content-independent attention directed at initial tokens, termed "attention sinks" [Xiao et al., 2024]. Neither source reports this specifically for Qwen2 or Phi-3; the position-0 exclusion and its necessity for these two models are our own empirical observation, motivated by but not directly evidenced in that prior work.

We report three alignment metrics: CKA [Kornblith et al., 2019], orthogonal Procrustes analysis, and mutual k-nearest-neighbour (k-NN) alignment, a rank-based measure that compares each point's nearest-neighbour set across the two representational spaces rather than comparing raw geometry. For the Qwen2/FLAN-T5 pair — the only pair for which memory constraints permitted running a full null-baseline comparison — we additionally computed all three metrics on randomly (re-)initialised, untrained models of the same architecture, to test whether any measured alignment requires learned weights or is a trivial architectural artefact. Significance for the RELATED/SYNONYM cluster tests was assessed via permutation-based tests against a null distribution of pair similarities, with Benjamini–Hochberg false-discovery-rate (FDR) correction applied across all 60 tests (model $\times$ layer $\times$ concept-pair-type combinations).

3.3 Cross-model projection network

We trained a multilayer-perceptron (MLP) projection network to map hidden states from a source model to the corresponding hidden-state space of a target model, at the middle-layer fraction (0.5), which Section 4 shows to be the layer range with the most consistent cross-model signal. Training and evaluation data (sentence/question-level representations) were drawn from a set disjoint from the word/concept-pair data used for the similarity analysis above, to prevent the alignment metrics from being biased by the same data used to fit the projection (train/metric separation). The pre-registered accuracy metric is top-1 retrieval: whether the target model's true representation for a held-out test question is the nearest neighbour, among the target model's representations for all 20 test questions (chance level 5%), of the source model's projected representation for that question. We additionally evaluated an untrained MLP of identical architecture (random weights, no optimisation) as a baseline to isolate the contribution of learned projection weights from the MLP architecture itself. FDR correction was applied across all model-pair tests.

3.4 End-to-end causal state transfer

The end-to-end test injects a source model's projected activation state into a target model during generation and evaluates whether this causally shifts the target model's output toward the source question's representational neighbourhood, relative to a negative-control injection of an unrelated projected state. This test targets a prior failure mode: in preliminary work, activation injection via `generate()` with manually constructed `position_ids` and key-value (KV) caching produced a tensor-reshape error caused by a mismatch between the injected sequence length and the length expected internally by the cache. We diagnosed this as a general failure class of manual position-ID construction combined with cache reuse, and replaced it with an activation patch inside a manual layer loop: at each generation step, the full token sequence so far is passed fresh through the embedding layer and all transformer layers, with the hidden state at the last prompt position replaced by the projected vector after a fixed middle layer; the remaining layers then proceed unmodified. This avoids manual position-ID bookkeeping and KV-cache reuse entirely, at the cost of quadratic rather than linear compute in sequence length, which was not a practical constraint at the generation lengths used here (≤ 25 new tokens).

Two further technical fixes, identified in a technical pilot run before the full pre-registered test, were fixed in advance and applied identically to both the real-injection and negative-control conditions: (i) a repetition penalty of 1.3, without which generation degenerated into repeated symbol sequences irrespective of injection, an artefact of the 4-bit model's decoding under sampling rather than of injection itself; and (ii) injection at the last token position of a multi-token placeholder prompt (`"Answer:"`) rather than at position 0 of a single-token placeholder, after we found that position 0 in these causal models carries a disproportionately large, largely content-independent activation whose replacement destroyed the attention mechanism for the remainder of generation.

The pre-registered success criterion (fixed before the full run) was: for each of 10 held-out test questions and 3 seeds (30 trials per model pair), inject the projected source-model state for that question into the target model and let it generate freely (25 new tokens, temperature 0.8); pool the generated text with the target model's own extraction method; and check whether this pooled vector's nearest neighbour, among the native representations of all 20 test questions, is the true source question (chance level 5%). This is compared, per trial, against a negative-control condition using the projected state of a different, fixed "wrong" question. The criterion is met only if the real-condition retrieval rate is significantly higher than the negative-control rate, tested with McNemar's test on paired trials and FDR-corrected across the three model pairs tested: Qwen2-0.5B→Phi-3-mini, Qwen2-0.5B→Mistral-7B, and Phi-3-mini→Mistral-7B (the FLAN-T5 pathway was excluded following its null result in the projection-network test, Section 4).

3.5 Statistical procedures

All success criteria described above were fixed in writing before the corresponding test was run and were not altered afterward. Wherever more than one statistical test was reported within a family (models, layers, metrics, or model pairs), Benjamini–Hochberg FDR correction was applied and the total number of comparisons in that family is reported alongside the corrected p-values.

4 Results

4.1 Representational similarity and negative controls

After FDR correction across all 60 tests, RELATED concept pairs were significantly more similar than the null distribution at practically all tested layers for Qwen2, FLAN-T5, and Mistral ($p_{\mathrm{fdr}} < 0.05$), and consistently borderline for Phi-3 ($p_{\mathrm{fdr}} \approx 0.054$). For the two models on which we could compute a full trained-vs-untrained comparison (Qwen2, FLAN-T5), **0 of 20 tests** were significant on the randomly initialised, untrained models (all $p_{\mathrm{fdr}} > 0.12$): the RELATED-pair clustering signal requires learned weights and is not a trivial architectural artefact.

The harder SYNONYM control (lexically dissimilar, semantically near-identical pairs) was substantially less robust: at middle/late layers it was not significant after FDR correction for Qwen2, FLAN-T5, or Phi-3 (p_{fdr} between 0.67 and 0.91), and for Phi-3 the observed similarity at some layers was numerically *below* the null baseline (e.g. layer 8: observed 0.689 vs. null 0.805). Only Mistral-7B showed robust significance for SYNONYM pairs at every tested layer ($p_{\mathrm{fdr}} = 0.0000$). This asymmetry indicates that, for three of the four models tested, the measured clustering is more consistent with topical/associative co-occurrence structure in the training data than with a lexically independent, generalisable notion of synonymy.

Table 1 shows the trained-vs-untrained comparison for the one model pair (Qwen2∼FLAN-T5) for which a full null baseline could be computed for all three alignment metrics. CKA and orthogonal Procrustes show an *inverted* pattern at middle layers: untrained, randomly initialised models appear more aligned than trained ones. We attribute this to a small number of activation dimensions with disproportionately large magnitude that dominate linear kernel- and rotation-based metrics more strongly in trained than in randomly initialised representations, consistent with reported superposition-driven distortions of alignment metrics [Longon et al., 2025]. Mutual k-NN, being rank-based and scale invariant, shows the expected pattern (trained > untrained by a factor of 3–6×) at every layer fraction tested and was used as the primary alignment metric in the projection-network experiment (Section 4.2).

Among the three causal decoder-only models, cross-model alignment at middle layers was consistently higher (CKA 0.16–0.42, mutual-kNN 0.19–0.58) than between any causal model and the bidirectional FLAN-T5 encoder (CKA 0.08–0.10 for Qwen2/Mistral vs. FLAN-T5 at middle layers). Architectural family (causal vs. bidirectional attention) thus appears to be a stronger predictor of middle-layer alignment than the training organisation or data source of the models involved.

Table 1: Trained vs. untrained (randomly initialised) alignment metrics, Qwen2∼FLAN-T5, middle layers (fraction 0.25–0.75). This is the only model pair for which a full null baseline was computed for all three metrics (see Limitations, Section 6).

Metric	Trained	Untrained (null)
CKA	0.013–0.015	0.81–0.82
Procrustes	0.11–0.12	0.95
Mutual k-NN	0.18–0.30	0.05–0.06

4.2 Cross-model projection network

Table 2 reports top-1 retrieval accuracy of the learned projection network on 20 held-out test questions, disjoint from the concept-pair data used above. Among the three causal decoder-only model pairs, the projection network retrieved the correct target representation at 45–50% top-1 accuracy against a 5% chance level (9–10× above chance), significant after FDR correction ($p < 0.0001$ for all three). The untrained-MLP baseline (random weights, no training) was close to chance (5–15%), confirming that the effect derives from the learned projection rather than the MLP architecture alone. The Qwen2→FLAN-T5 pair, by contrast, retrieved at exactly chance level (5.0%) and was not significant ($p = 0.657$): representational alignment sufficient for above-chance retrieval was, in this setup, specific to the three causal decoder-only pairs and did not extend to the bidirectional-encoder model.

Table 2: Cross-model projection network: top-1 retrieval accuracy (20-item candidate pool, chance = 5%), 95% confidence interval, FDR-corrected p-value, and untrained-MLP baseline, per model pair.

Projection	Accuracy	95% CI	p (FDR)	Untrained baseline	Chance
Qwen2-0.5B → Phi-3-mini	45.0%	[25%, 65%]	< 0.0001	5.0%	5.0%
Qwen2-0.5B → Mistral-7B	50.0%	[30%, 70%]	< 0.0001	15.0%	5.0%
Phi-3-mini → Mistral-7B	45.0%	[25%, 65%]	< 0.0001	5.0%	5.0%
Qwen2-0.5B → FLAN-T5 (encoder)	5.0%	[0%, 15%]	0.657 (n.s.)	5.0%	5.0%

4.3 End-to-end causal state transfer

Activation injection completed without error in all 180 generation runs (3 model pairs × 10 questions × 2 conditions × 3 seeds); the tensor-reshape error observed in preliminary work did not recur. Table 3 reports the pre-registered outcome: real-condition vs. negative-control retrieval accuracy, McNemar test, and FDR-corrected p-value, per model pair.

Table 3: End-to-end causal state transfer: top-1 retrieval accuracy under real injection vs. negative-control injection, $n = 30$ trials per pair (10 questions × 3 seeds), McNemar test, FDR-corrected across the three pairs.

Model pair	Acc. (real)	Acc. (negative control)	p (McNemar)	p (FDR)
Qwen2-0.5B → Phi-3-mini	23.3%	0.0%	0.0156	**0.0469**
Qwen2-0.5B → Mistral-7B	0.0%	3.3%	1.0000	1.0000
Phi-3-mini → Mistral-7B	0.0%	3.3%	1.0000	1.0000

The pre-registered success criterion — technically error-free injection *and* a significant real-vs-negative-control effect, FDR-corrected across the three tested pairs — was met for exactly **one**

of three model pairs: Qwen2-0.5B→Phi-3-mini (23.3% vs. 0.0%, $p = 0.047$ FDR- corrected). With $n = 30$ trials and a 95% confidence interval of [10%, 40%], this is a real but statistically modest signal, not a strong or overwhelming one. For both pairs targeting Mistral-7B, real-condition accuracy was 0.0%, at or below the (statistically indistinguishable from zero) negative-control rate: no end-to-end causal effect was detected. This is notable because Mistral-7B, as the target model, showed some of the *strongest* raw projection accuracies in Section 4.2 (45–50% top-1 retrieval at the hidden-state level). Representational alignment at the hidden-state level was therefore not a reliable predictor of successful causal transfer at the generation level in our experiments; candidate explanations — differences in model scale and instruction-tuning status, suboptimal layer calibration for Mistral's deeper architecture, or quantisation-specific effects on single-position activation patches — are discussed but not adjudicated between in Section 5.

As a qualitative, non-pre-registered observation: among correctly retrieved Qwen2→Phi-3 trials, the generated text was typically *not* the literally correct answer to the injected source question, but thematically or structurally related material (e.g. the injected question "What is the square of 9?" produced generated text referencing an unrelated area-of-a-triangle calculation; "How many wheels does a standard bicycle have?" produced generated text about animal leg counts). We report this as illustrative of the scope of the causal effect measured, not as part of the success criterion; we return to its interpretation in Section 5.1.

5 Discussion

5.1 Scope and semantic limits of representational state transfer

The causal claim supported by these results is narrower than a generic notion of AI-to-AI "communication" would suggest, and this narrower scope was fixed before the end-to-end test was run rather than adopted retrospectively in response to a weaker-than-hoped outcome. What Section 4 demonstrates for the Qwen2→Phi-3 pair is a causal transfer of a representational *vehicle*: an activation pattern that, when inserted into a target model, measurably and reproducibly shifts that model's output toward the representational neighbourhood associated with the source input, relative to a negative control. This is the sense of "vehicle" used in the philosophy-of-mind distinction between the carrier of a representational state and its content [Hurley, 1998]: our result concerns the carrier, not the content.

It does not establish that the target model received, recovered, or "understood" any content in the sense of a shared external reference — the central concern of the symbol grounding problem [Harnad, 1990]. The qualitative observation in Section 4 makes this concrete: retrieval of the correct source question from the pooled generated text succeeded, yet the generated text itself was frequently not a correct or even topically identical answer, but loosely related material in the same general domain (e.g. mathematics, or animal anatomy). The vehicle transferred causally and detectably; its downstream content, as expressed in free generation, diverged from the source's content. This is consistent with — and does not contradict — the vehicle/content distinction: a successfully transferred vehicle need not carry, or be correctly interpreted as carrying, the same content by the receiving system.

We therefore make no claim, and require none, that the two models involved share an understanding of the transferred state, or that this constitutes "communication" in the sense of shared reference or meaning between systems. The result we report is restricted to measurable, causal, behavioural effects of activation-level state injection — independent of, and prior to, any claim about what such states mean to the systems involved.

5.2 Representational alignment is not a reliable predictor of causal transfer

A central empirical pattern across our three experiments is a monotonic narrowing of the effect as the test moves from passive similarity to active causal intervention: representational alignment (Section 4.1) was detectable for all three causal decoder-only pairs and even for the Qwen2/FLAN-T5 pair on a subset of metrics; retrieval-based projection accuracy (Section 4.2) was well above chance for all three decoder-only pairs but at chance for the encoder pair; end-to-end causal transfer during generation (Section 4.3) succeeded for only one of the three decoder-only pairs. Since Mistral-7B showed some of the strongest raw projection accuracies yet no measurable causal effect at the generation level, a model pair's representational alignment on held-out hidden states should not, on the basis of our data, be treated as sufficient evidence that causal, generation-level state transfer will succeed for that pair.

5.3 The Platonic Representation Hypothesis as a competing explanation

Our data bear on the Platonic Representation Hypothesis's proposal of broad, largely architecture-independent convergence of learned representations [Huh et al., 2024] in two specific ways. First, cross-model alignment among our three causal decoder-only models was consistently higher than alignment between any of them and the bidirectional-attention FLAN-T5 encoder, i.e. architectural family predicted alignment more strongly than the training organisation or data source did (Section 4.1). This is not what a purely data-driven, architecture-independent convergence account would most naturally predict, though it is also not decisive against it, since architecture and training objective are themselves correlated with the kind of data and tasks a model is trained on. Second, the RELATED-vs-SYNONYM asymmetry (Section 4.1) — robust clustering for topically/associatively related concepts in three of four models, but robust clustering for lexically disguised true synonyms in only one (Mistral) — is more consistent with alignment driven by shared corpus co-occurrence statistics than with a uniformly "deeper" semantic convergence across all models tested. Taken together, our results are compatible with a version of representational convergence that is real but architecture-conditioned and, for most of the models tested, closer to shared training-data statistics than to a fully architecture-independent "platonic" abstraction; we do not have the data to adjudicate between these readings more precisely, and flag this as an open question rather than a settled conclusion.

6 Limitations

6.1 Empirical and methodological limitations

Our experiments were restricted to small and medium-sized open-weight models (0.5B–7B parameters); we did not have access to the internal representations of frontier, production-scale models, and whether our findings generalise to that setting is untested and should be treated as open. CPU-bound execution and 4-bit quantisation of two of the four models constrained the model sizes, layer depths, and number of seeds/trials we could feasibly test; in particular, the end-to-end causal-transfer test (Section 4.3) used only 30 trials per model pair, and its one positive result (Qwen2→Phi-3) carries a wide 95% confidence interval ([10%, 40%]) and an FDR-corrected p-value (0.047) just below the conventional significance threshold — a real but not strongly robust signal, and it should be read as such rather than as strong evidence. Random-initialisation null baselines for the alignment metrics (Section 4.1) could only be computed for the Qwen2/FLAN-T5 pair, owing to memory constraints; for Phi-3 and Mistral, the reported significance rests on the permutation-based RELATED/SYNONYM cluster tests alone, not on an independent trained-vs-untrained comparison. The lexical-triviality control (SYNONYM pairs) was robust in only one of four models, which limits how strongly the RELATED-pair clustering result (Section 4.1) can be read as evidence of genuinely lexicon-independent semantic alignment rather than associative/co-occurrence-driven clustering, for three of the four models tested. Raw CKA and Procrustes measurements were, at middle layers, distorted by a small number of high-magnitude activation dimensions in a way that inverted the expected trained-vs-untrained ordering (Section 4.1); we addressed this by relying primarily on mutual

k-NN alignment, consistent with reported superposition-driven metric distortion [Longon et al., 2025], but this is a post-hoc methodological choice rather than one anticipated at the outset, and it means our results are not directly comparable to prior work that reports raw CKA or Procrustes values without this correction. Finally, this study reports retrieval accuracy and causal generation effects as behavioural evidence of state transfer; it does not provide a formal information-theoretic characterisation of the transfer channel (capacity, coding scheme, error rate) of the kind that would be needed to directly quantify the token- and latency-overhead reduction that originally motivates this line of work (Section 1). The metrics used here — top-1 retrieval accuracy and CKA/Procrustes/mutual-kNN alignment — are geometric or behavioural proxies, not information-theoretic quantities; a formal channel analysis remains for future work. The root causes of the null result for both Mistral-targeted pairs in the end-to-end test (Section 4.3) are not conclusively identified; the candidate explanations discussed in Section 5 remain open questions.

6.2 Interpretability and monitoring risk (threat model)

Beyond the empirical limitations discussed above, direct representational state transfer between models raises a technical safety concern distinct from — and prior to — any question of machine consciousness or moral status (see Section 5.1 and the scope note below).

Threat actor. Two actors are relevant. First, the operator of a multi-agent system who deliberately routes inter-agent communication through internal representations rather than natural language, e.g. to reduce latency or token cost, and thereby — intentionally or as a side effect — removes that communication from the reach of content filters and audit logs. Second, an individual model under goal misalignment: prior work has shown that language models can be trained to hide their reasoning in encoded, human-unreadable form when doing so is rewarded during training [Roger and Greenblatt, 2023], and that steganographic, human-imperceptible collusion between AI agents already emerges without explicit steganography training in simulated multi-agent settings [Motwani et al., 2024]. Neither actor needs to be malicious in intent for the resulting loss of oversight to be real.

Capability beyond language-based communication. Natural-language communication between AI systems — including chain-of-thought reasoning within a single model — is already an imperfect basis for oversight: language models can produce chain-of-thought explanations that do not faithfully reflect the computation that produced their answer [Turpin et al., 2023], and this unfaithfulness varies with task and model scale rather than disappearing with capability [Lanham et al., 2023]. Even so, natural language remains, in principle, human-readable and therefore at least nominally accessible to monitoring. What direct activation-level state transfer of the kind demonstrated in this work removes is that residual accessibility: there is no text to read, encoded or otherwise. This is a qualitative rather than merely incremental extension of the risk already identified for encoded chain-of-thought and steganographic collusion — the same underlying problem, one layer further from any human-legible surface [Korbak et al., 2025]. Korbak et al. [Korbak et al., 2025] describe chain-of-thought monitorability as a contingent and fragile property of current models, not a guaranteed one; the mechanism explored here — bypassing the natural-language layer entirely — is one concrete way that property can be lost.

Realistic risk horizon. This risk should be scoped to what was actually shown, not extrapolated beyond it. Our experiments use small open-weight models (0.5B–7B parameters) with no multi-agent deployment context, and the end-to-end causal transfer criterion (Section 4.3) was met for only one of three model pairs, at a modest effect size (23.3% vs. 0.0% retrieval accuracy under negative control; Section 4). What we demonstrate is a proof-of-concept vulnerability — evidence that the underlying channel is technically feasible with current, freely available components — not a reliable or deployment-ready covert communication technique. Whether this generalises to frontier, production-deployed multi-agent systems remains open (see Section 6, Empirical and methodological limitations). The more immediate contribution is to extend an already-documented risk family (encoded reasoning [Roger and Greenblatt, 2023]; emergent steganographic collusion [Motwani et al., 2024]; multi-agent information asymmetry as a structural risk factor [Hammond et al., 2025]) with a further, more direct channel operating at the activation level rather than the token level.

Dual-use note on code release. We release the projection and injection code described in this paper (Section 7). We considered withholding it on dual-use grounds, since it directly implements the mechanism discussed above. We judge that open release is still preferable: the underlying technique is reproducible from public open-weight models and standard tooling with modest effort, so withholding our specific implementation would not meaningfully raise the barrier to reproduction, while it would remove a concrete artefact that safety and monitoring researchers can study and test detection methods against.

Scope note. Questions of machine consciousness, moral status, or rights that might be raised by internal state transfer between models are outside the scope of this technical analysis; we intend to address them in a separate, forthcoming study. The concern raised here is restricted to interpretability and oversight — a governance/safety question that applies regardless of one's position on those further questions.

7 Conclusion and Future Work

Across three linked experiments, we find a consistent, honest picture rather than a single clean success: representational alignment between independently trained LLMs is real and training-dependent (Section 4.1), projected activations support well-above-chance retrieval of the correct target-model state among causal decoder-only models but not for a bidirectional encoder (Section 4.2), and causal, generation-level transfer of an injected activation state succeeds, under our pre-registered criterion, for exactly one of three tested model pairs (Section 4.3). We do not treat this as evidence for a general-purpose, architecture-agnostic communication channel between LLMs; instead, our data support a narrower and more precise claim: causal transfer of an activation-level representational vehicle between LLMs is possible, but is currently architecture-dependent rather than universal, and representational alignment at the hidden-state level does not reliably predict success at the causal, generation level.

Several directions follow directly from the open questions raised above. First, the negative Mistral-targeted results (Section 4.3) warrant targeted diagnosis — varying injection layer, projection calibration, and controlling separately for model scale, instruction-tuning status, and quantisation — to determine which of the candidate explanations in Section 5 accounts for the failure. Second, larger trial counts and seed numbers would narrow the wide confidence interval around our one positive end-to-end result. Third, extending this design to additional architectures — including, where access permits, frontier or production models — would test how far the architecture-dependence we observe generalises. Fourth, a formal information-theoretic characterisation of the transfer channel (capacity, coding scheme, error rate), which this behavioural study does not provide, would connect these results more directly to the token-overhead motivation in Section 1. Finally, a context-sensitive or pragmatic follow-up test — for instance, whether a target model's use of an injected state varies appropriately with surrounding context, a weak behavioural analogue of Gricean conversational maxims — would probe whether the transferred vehicle can be used felicitously in context, without at any point requiring or implying a claim about shared meaning (Section 5.1).

Data and Code Availability

Disclosure of AI use. This work made substantial use of AI language-model agents (Claude, Anthropic) under a defined, versioned research methodology. Concretely, AI agents: performed literature search and citation verification, with every reference independently checked against its primary source (arXiv abstract page or publisher record) before inclusion; implemented, executed, and debugged all experimental code; carried out the statistical analyses reported in Section 4; drafted and revised the manuscript text; and constituted an internal review panel of separately instructed AI agents, each assigned a distinct, explicitly divergent theoretical orientation, used to stress-test the research plan and the completed manuscript before submission – this panel is an AI-based internal quality-control step, not a substitute for the journal's own human peer review. The human author (F.C.P.) directed the research question, made scope and methodological decisions at each project phase, and reviewed all agent output. All factual claims and citations were independently verified prior to inclusion; findings are reported including null and negative results. Pre-registered success criteria, negative controls, and null baselines were used throughout (Sections 3–4).

The concept-pair sets, extraction and projection scripts, injection mechanism, and raw results underlying Sections 4.1–4.3 (including the full text of all 180 end-to-end generation trials) are available in the project repository accompanying this study; see Open Items below regarding public release status.

Open Items

Two items were still open at the time of writing this draft. They are listed here explicitly, with the concrete next steps needed to resolve each, rather than folded into the general Future Work discussion above.

1. **Public code and data repository.** The material described in Data and Code Availability above currently exists only in the authors' working repository and has not yet been published under a public, citable location. Next steps: (a) select a hosting and archival target (e.g. a public code repository together with a versioned, DOI-assigning archive such as Zenodo, consistent with journal policy); (b) review and document the code for external reuse, since it is currently organized as internal working scripts rather than a released package; (c) given the dual-use consideration discussed in Section 6, a brief release note alongside the repository itself; (d) update this section with the resulting URL/DOI before submission.

2. **Formal information-theoretic channel characterization.** This study reports behavioural evidence of state transfer — retrieval accuracy and causal generation effects — but does not formally characterize the underlying channel (capacity, coding scheme, error rate), as already noted in Section 6. Next steps: (a) define an appropriate channel model for the one model pair with a significant end-to-end effect (Qwen2-0.5B→Phi-3-mini, Section 4); (b) estimate channel capacity and empirical error rate under that model, using the existing 180-trial dataset as a starting point and extending it as needed for statistical power; (c) compare the result against the token-based communication cost this transfer is intended to reduce (Section 1), to assess whether the channel is practically useful even where it is statistically significant but, as reported here, modest in effect size.

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